University of São Paulo

Institute of Mathematical and Computational Sciences (ICMC)

Asymptotics of Partition Functions on Coulomb Gases

Annual Scientific Report for the of Master Regular Program, supported by São Paulo Research Foundation.

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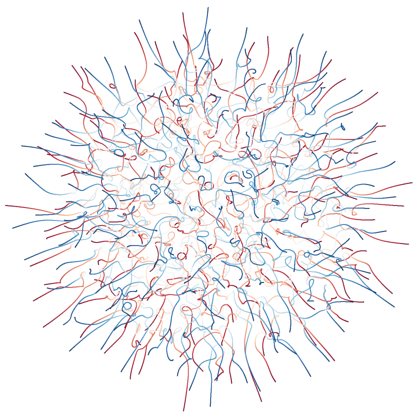

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General Information on The Project

- Project Title:

  Asymptotics of Partition Functions on Coulomb Gases

- Advisors Name:

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- Institution:

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- Research Team:

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- Grants Number:

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- Period of validity:

  01/08/2024 a 31/12/2025

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  01/08/2024 a 20/03/2025

Abstract

The theory of Random Matrices finds its way in a diverse range of areas of Mathematics and Physics, and in particular to statistical mechanics. In the study of the spectra of eigenvalues of some random matrices ensembles, a physical analogy regarding thermodynamical systems becomes evident and the use of physical motivations in its study is of great importance. An example of such motivation is the study of the partition function, which holds in thermodynamics a great deal of information on the system at hand. Recent developments have been intensely unraveling new critical phenomena in random matrix theory, through a deep understanding of asymptotic expansions of the associated partition functions in the thermodynamic limit. The main goal of this work is to understand these developments in the context of normal random matrices, their relevance and implications in the theory of random matrices and beyond.

1 Realizations in the Period

1.1 Masters

During the period referred by this report, the first semester of the regular master program, the student has completed the courses listed in the table. In addition to that, he has also taken a course in the Summer Course in the Institute of Pure and Applied Mathematics (IMPA). By the time of submission activities for the second semester have started and the courses in which he is enrolled are also listed below.

Course Name	Course Code	Grade	Semester
Smooth Manifolds	SMA5781-21/3	A	2 - 2024
Ordinary Differencial Equations	SMA5802-17/2	B	2 - 2024
Algebra	SMA5738-17/4	-	1 - 2025
Measure and Integration	SMA5801-13/4	-	1 - 2025
Pedagogical Preparation	SMA5839-19/8	-	1 - 2025

1.2 Research

During the months referred by this report the effort was spent working on the regular courses of the masters program, already cited in the table above. Because of that, activities related to the research topic are limited.

1.2.1 Scientific Introduction

A random matrix is a matrix in which the entries are random variables, not necessarily independent nor equally distributed. The algebraic-geometric properties in a matrix, such as its natural multiplication and spectral decomposition, makes this representation especially useful. Naturally, many complex systems use a matrix representation. Correlation matrices and operators, especially in physics, are some of many major reasons why this representation is important. Studying these matrices we can deduce properties from the system of interest, either that being the eigenvalues and eigenvectors of operators describing an atomic nuclei [1] or the description of market shares in highly correlated systems [2, Chapter 2]. Either way, using an random matrix approach is relevant by the same reason random variables proved themselves crucial: it unlocks relevant statistical description of phenomena and systems.

In the research proposed here, we will focus on the study of *normal random matrices*, which consists of the space of normal matrices, equipped with a proper probability distribution. In our case, such probability distribution is given by a Gibbs-type measure, which we also impose to be invariant under the action of the unitary group. All in all, such invariance imposes a trivialization of the eigenvectors, which turn out to be simply Haar-distributed on the unitary group. In turn, all the relevant information on the matrix model is contained in their eigenvalues, and their joint probability density function is explicitly given by

$$p(\lambda_1, \lambda_2, \ldots, \lambda_N) = \frac{1}{Z_{N,\beta}} \mathrm{e}^{-\beta_N \mathcal{H}_N(\vec{\lambda})}, \quad (\lambda_1, \ldots, \lambda_N) \in \mathbb{C}^N \tag{1.1}$$

where $Z_{N,\beta}$ is the canonical partition function, such that p is a probability density with respect to the Lebesgue measure on $\mathbb{C}^N$. The factor $\beta_N = \beta N^2$ is though of as the inverse temperature and the factor $\mathcal{H}_N$, called the Hamiltonian, is given by

$$\mathcal{H}_N(\vec{\lambda}) = \frac{1}{N}\sum_{i=1}^{N} V(\lambda_i) + \frac{1}{N^2}\sum_{i<j} \log \frac{1}{|\lambda_i - \lambda_j|}, \quad (\lambda_1, \ldots, \lambda_N) \in \mathbb{C}^N$$

where $V : \mathbb{C} \to \mathbb{C}$ is a suitably regular function, acting as a confining potential on the eigenvalues $\lambda_1, \ldots, \lambda_N$.

Considering such a measure is natural to make an analogy to the well know Coulomb Gas. Under appropriate conditions, the Coulomb Gas is a Gibbs-Boltzmann probability measure given in $(\mathbb{R}^d)^N$. This measure p_N models an interacting gas of charged particles, at $x_1, x_2, \ldots, x_N \in \mathbb{S}$ of dimension d in $\mathbb{R}^n$ ambient space, under the influence of an external potential. Its measure its given by

$$p_N(x_1, x_2, \ldots, x_N) = \frac{e^{-\beta N^2\, \mathcal{H}_N(x_1, x_2, \ldots, x_N)}}{Z_{N,\beta}}, \tag{1.2}$$

where

$$\mathcal{H}_N(\vec{x}) = \frac{1}{N}\sum_{i=1}^{N} \mathrm{V}(x) + \frac{1}{2N^2}\sum_{i\neq j} \mathrm{g}(x_i - x_j)$$

is the Hamiltonian or energy of the system and $\mathrm{g}(x_i - x_j)$ is the Coulomb Kernel of interaction. In this analogy, eigenvalues $\lambda_1, \ldots, \lambda_N$ may be seen as particles under the influence of the Gibbs law, and thermodynamical arguments may be employed to describe the configurations of eigenvalues of random matrices. With this in mind we turn to the partition function. In the beginning of its book, Feynman states [3] that the key principle of the statistical mechanics is that the probability that a system attains a state with energy E, in equilibrium, is given by a function $\frac{1}{Z}\mathrm{e}^{\frac{-E}{kT}}$ where Z is the partition function. In a more general sense there is a relation between the partition function and the free energy of the system, that, in itself, relates to the entropy. Knowing well the partition function is a way to describe with great detail the macro-properties of such a system.

There is a great effort in the community of random matrix theory to study expansions of the partition function of systems such of Coulomb Gases. Recently, some advance has been made in the works of Sug-Soo Byun et al. [4]. They derived large-N asymptotic expansions up to the $O(1)$-terms for $Z_{N,\beta}$ when V is a radially symmetric potential, in the form

$$Z_{N,\beta=2} = -I^V N^2 + \frac{1}{2} N \log N + E^V N + G \log N + F^V + o(1), \quad N \to \infty.$$

In such expansion, for a large class of potentials V the first term I^Q was known for a long time, being given by the energy of the associated equilibrium measure. The coefficient $1/2$ of the order $N \log N$, as well as the entropy term E^V, were known from recent work of Leblé and Serfaty [5], also for rather large classes of V. The remaining terms in this expansion are new, and they are computed exploring the radially symmetry imposed on the potential. Strikingly, the term G appears to be universal in V, depending solely on the connectivity of the limiting spectrum of eigenvalues. More precisely, they noticed that G depends on whether the limiting spectrum is an annulus or a disc which, surprisingly, seems to not have been observed in the vast previous literature on the matter.

As was said, the study of the partition function is a major way to describe thermodynamical systems such that of a Coulomb Gas. With these new developments it is expected that many systems could be better understood and described by the asymptotic given. In that way this poses a great opportunity for development in the field. More than that, by the great connection that the field of random matrices holds with many other fields is possible that by the study of this work some suggestions on the behavior of many other related problems become plausible.

1.3 Cronogram

This M.D. research has started in August 2024 and is planned to run for the usual time for the Masters Program, that is, 24 months. The project is being run with weekly meetings with the supervisor, and planned sporadic seminar expositions to the other members of the research team associated with the JP Research Grant and correlated groups. We divide our activities in five parts

1. Master Courses: Execution of the regular courses provided by the program to work out general and specific mathematical tools for the research and academical development;

2. Concepts: the specific study of the mathematical toolkit for Statistical Mechanics and fields related to the research topic;

3. Main Article: Exploring and understanding results of the article that motivates our research proposal, [4];

4. Exploration: Using the tools developed for the explorations of possible new results and possibilities in the field;

5. Master Thesis: Writing te Master Thesis, as required by the program.

The first three quarters have been completed as planned.

Steps	**Quarters**							
	1	2	3	4	5	6	7	8
1. Master Courses	x	x	x	x	x	x		
2. Concepts			x	x	x	x		
3. Main Article				x	x	x	x	
4. Exploration					x	x	x	
5. Masther Thesis					x	x	x	x

Table 1.1: Activities planned, the goals are as described above. Red color indicates the focus of the quarter that may be divided between activities for some crucial quarters. Green is for the quarters that have been completed as planned

2 Attendance in Scientific Events

As it has only been one semester on the program and the main focus was the regular courses, the student has not yet taken part in a scientific event for presentation purposes. He has however, in addition to the participation in the Summer School of IMPA, taken part in a MasterClass on Random Permutations by Mattia Cafasso and Sofia Terracome on Angers, France, offered by the Centre de Mathematiques Henry Lebesgue. For this event it was used the Technical Reserve of the Scholarship as almost two daily stipends on international scientific event (limited by available resource).

Event	Where	Date	Type	Presented	Tech. Reserve
Summer School	IMPA	05/01-28/02/25	In person	No	No
MasterClass	Angers	16-20/12/24	In person	No	Yes

3 Other prepared or submitted works

During the last semester an article which I co-authored have been submitted for publication and is available in Arxiv for public consultation at Simulating Simple Random Walks With a Deck of Cards (https://arxiv.org/abs/2410.00831). This work is the result from a short term scientific research program done in ICMC in collaboration with the other authors in the paper. The program consisted of a month and a half of intense discussions and development of the research topic with other students and researches from many institutes and backgrounds. The topic and questions were developed during the time in the program.

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[4] BYUN, S.-S.; KANG, N.-G.; SEO, S.-M. Partition functions of determinantal and pfaffian coulomb gases with radially symmetric potentials. *Communications in Mathematical Physics*, Springer Science and Business Media LLC, v. 401, n. 2, p. 1627–1663, mar. 2023. ISSN 1432-0916. Disponível em: <http://dx.doi.org/10.1007/s00220-023-04673-1>.

[5] LEBLé, T.; SERFATY, S. *Large Deviation Principle for Empirical Fields of Log and Riesz Gases*. 2017.

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Programa:	Matemática
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Janus

Data do Depósito do Trabalho:

Título do Trabalho:

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Histórico de Ocorrências: Primeira Matrícula em 30/07/2024

Aluno matriculado no Regimento da Pós-Graduação USP (Resolução nº 7493 em vigor a partir de 29/03/2018).

Última ocorrência: Matrícula Regular em 20/01/2025

Sigla	Nome da Disciplina	Início	Término	Carga Horária	Cred.	Freq.	Conc.	Exc.	Situação	Pres.
SMA5781-21/3	Variedades Diferenciáveis	12/08/2024	29/11/2024	150	10	100	A	N	Concluída	S
SMA5802-17/2	Equações Diferenciais Ordinárias	13/08/2024	29/11/2024	120	8	100	B	N	Concluída	S
SMA5738-17/4	Álgebra	10/03/2025	17/07/2025	120	0	-	-	N	Pré-matrícula deferida	S
SMA5801-13/4	Medida e Integração	11/03/2025	17/07/2025	120	0	-	-	N	Pré-matrícula deferida	S
SMA5839-19/8	Preparação Pedagógica	13/03/2025	17/07/2025	60	0	-	-	N	Pré-matrícula deferida	S

	Créditos mínimos exigidos Para depósito da dissertação (Totais/Presenciais)	Créditos obtidos Totais/Presenciais
Disciplinas:	48 / 29	18 / 18*
Estágios:		
Total:	48 / 29	18 / 18*

Créditos Atribuídos à 48

*Créditos obtidos anteriormente a 01/01/2024 serão computados como presenciais de acordo com a Resolução CoPGr 8546, de

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A - Excelente, com direito a crédito; B - Bom, com direito a crédito; C - Regular, com direito a crédito; R - Reprovado; T - Transferência. Um(1) crédito equivale a 15 horas de atividade programada.

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